

PATHFINDER.

BY PATHFINDER LIDAR SOLUTIONS

REQUIREMENTS

Acceptance Criteria

97 Given / When / Then criteria, each independently testable and mapped to a test.

Project	Pathfinder Portal v2
Version	0.1 (Draft for review)
Date	2026-06-15
Owner	Pathfinder LiDAR Solutions

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Field	Value
Document	03 - Acceptance Criteria
Version	0.1
Date	2026-06-15
Status	Draft
Source of truth	docs/00-canonical-context.md
Companion	docs/02-user-stories.md

Intro

This document gives the acceptance criteria for every user story in `docs/02-user-stories.md`. Each criterion is written in Given/When/Then form and carries an ID `AC-<story>.<n>`. Each AC is independently testable and maps 1:1 to a Vitest/Playwright test `T-<AC>` in the forthcoming TDD plan (the canonical chain Epic -> US -> AC -> Test -> Sprint).

Criteria describe behavior and outcomes, never implementation. Per the documentation convention (research 04), each story carries 1 to 3 ACs; coverage spans the happy path plus at least one edge, permission, or error case in each epic. A trailing `[screenshot]` tag marks a UI-visible outcome that the TDD plan should assert with a Playwright visual snapshot.

E1 - White-Label Brand Layer

US-1.1 - Default Pathfinder brand

- **AC-1.1.1** Given no `BRAND` env var is set, When the app loads, Then the Pathfinder LiDAR brand name, logo, and dark-slate color tokens render. [screenshot]
- **AC-1.1.2** Given the default brand is active, When any component reads a brand color, Then it resolves from a CSS custom property and never from a hard-coded value.

US-1.2 - Drop-in brand swap

- **AC-1.2.1** Given `BRAND=els911` is set at build, When the app loads, Then the ELS911 product name, logo, and palette render in place of Pathfinder's with no component change. [screenshot]
- **AC-1.2.2** Given a brand profile is missing its logo asset, When the app loads, Then it falls back to the landmark placeholder and logs a warning rather than rendering broken.

US-1.3 - Ready-made ELS911 profile

- **AC-1.3.1** Given the ELS911 profile, When it is activated, Then its tokens resolve to primary red `#B22234` and navy `#002868` as ported from v1 tokens.css. [screenshot]

US-1.4 - Token-only component theming

- **AC-1.4.1** Given the component source tree, When it is scanned for color literals, Then no component file contains a hard-coded brand hex value.
- **AC-1.4.2** Given a brand token value is changed, When the app re-renders, Then every component referencing that token reflects the new value with no component edit. [screenshot]

US-1.5 - Brand continuity with marketing site

- **AC-1.5.1** Given the active brand profile, When a user moves from the marketing site to the logged-in product, Then the logo, primary color, and typography match. [screenshot]
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E2 - Identity, Roles & Access

US-2.1 - Email/password login

- **AC-2.1.1** Given a registered user, When they submit valid credentials, Then a JWT session is issued and they land on their scoped dashboard. [screenshot]
- **AC-2.1.2** Given a registered user, When they submit a wrong password, Then login is rejected with a generic error and no session is issued. [screenshot]

US-2.2 - Role-based access control

- **AC-2.2.1** Given a user with the client role, When they request a staff-only route, Then access is denied with a 403 and a permission screen is shown. [screenshot]
- **AC-2.2.2** Given a user with the admin role, When they request an admin route, Then the route renders.

US-2.3 - Org-scoped data isolation

- **AC-2.3.1** Given a client reviewer in Org A, When they request a facility belonging to Org B by ID, Then the request is denied and no Org B data is returned.

US-2.4 - Audit-logged authentication events

- **AC-2.4.1** Given audit logging is enabled, When a user logs in, logs out, or is denied access, Then an immutable audit entry records the actor, event, and timestamp.

US-2.5 - Forward-looking MFA/SSO

- **AC-2.5.1** Given the auth layer, When a user record carries an MFA-enabled flag, Then the login flow exposes the MFA step hook without breaking password-only users.
-

E3 - Org Hierarchy & Dashboards

US-3.1 - Navigate the org hierarchy

- **AC-3.1.1** Given an org with districts, facilities, and buildings, When a staff member drills from org to building, Then each level lists only its direct children. [screenshot]

- **AC-3.1.2** Given the hierarchy, When rendered, Then nesting is capped at the canonical three levels below org (district, facility, building).

US-3.2 – Breadcrumb switcher dropdowns

- **AC-3.2.1** Given a building page, When a user clicks a breadcrumb segment, Then a dropdown of sibling entities at that level opens. [screenshot]
- **AC-3.2.2** Given the breadcrumb dropdown, When the user selects a sibling, Then the view navigates to that sibling at the same hierarchy level.

US-3.3 – Roll-up dashboards

- **AC-3.3.1** Given a district with multiple facilities, When the operator admin opens the district dashboard, Then it shows an aggregated roll-up of facility and project status. [screenshot]

US-3.4 – Designed empty states

- **AC-3.4.1** Given a facility with no projects, When a client reviewer opens it, Then a designed empty state shows a sample preview and one clear first action. [screenshot]

US-3.5 – Favorites and saved views

- **AC-3.5.1** Given a building, When a staff member stars it, Then it appears in their favorites list and persists across sessions. [screenshot]

E4 – Project & Review Workflow

US-4.1 – Create a project under a building

- **AC-4.1.1** Given a building, When a staff member creates a project with a name and members, Then the project is saved with that `building_id` and an initial status. [screenshot]
- **AC-4.1.2** Given project creation, When no building is selected, Then creation is blocked with a validation error.

US-4.2 – Track project progress and status

- **AC-4.2.1** Given a project, When a client reviewer opens it, Then its current status and progress are displayed without any staff-only editing controls. [screenshot]

US-4.3 – Version review loop

- **AC-4.3.1** Given a staff member publishes V1, When publication completes, Then the client org's reviewer is notified and the version is marked awaiting review.
- **AC-4.3.2** Given client annotations on V1, When staff publish V2, Then V2 becomes current and V1's annotations remain tied to V1 for history. [screenshot]

US-4.4 – Record an approval

- **AC-4.4.1** Given a current version, When a client reviewer approves it, Then the approval is recorded against that version with actor and timestamp. [screenshot]

- **AC-4.4.2** Given a user without the client reviewer role, When they attempt to approve, Then the action is denied.

US-4.5 - Manage project membership

- **AC-4.5.1** Given a project, When an operator admin removes a member, Then that user loses access to the project on their next request.
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E5 - 2D Floorplan Annotation Engine

US-5.1 - View a floorplan PDF

- **AC-5.1.1** Given a floorplan PDF asset, When a staff member opens it, Then the PDF renders in the in-portal viewer at the correct page and scale. [screenshot]
- **AC-5.1.2** Given a corrupt or unsupported file, When opened, Then the viewer shows an error state instead of a blank canvas. [screenshot]

US-5.2 - Use the annotation tools

- **AC-5.2.1** Given the viewer, When a staff member selects each of the ported tools and draws, Then the corresponding annotation is created on the canvas. [screenshot]

US-5.3 - Persist annotations as JSON

- **AC-5.3.1** Given annotations drawn on a floorplan, When the page is reloaded, Then the annotations reappear exactly as drawn. [screenshot]
- **AC-5.3.2** Given saved annotations, When exported, Then the output is valid JSON round-trippable back into the same shapes.

US-5.4 - Layered per-audience annotations

- **AC-5.4.1** Given annotations tagged by audience layer, When a reviewer toggles the responder layer off, Then only non-responder annotations remain visible. [screenshot]

US-5.5 - Client annotation in the review loop

- **AC-5.5.1** Given a client reviewer viewing a draft, When they place an annotation on a room, Then it is saved against the current version and visible to staff. [screenshot]
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E6 - Safety Mapping & NFPA Export

US-6.1 - Place NFPA 170 markers

- **AC-6.1.1** Given the NFPA 170 catalog, When a staff member places an AED marker, Then it renders with the standardized NFPA 170 symbol. [screenshot]
- **AC-6.1.2** Given the marker tool, When a user attempts a freehand symbol, Then only catalog symbols are placeable and freehand symbol creation is not offered.

US-6.2 - Auto-generated NFPA legend

- **AC-6.2.1** Given placed markers, When the legend generates, Then it lists one vertical row per marker type with hallway chips and no duplicate rows. [screenshot]

US-6.3 - North-locked printable map render

- **AC-6.3.1** Given a map for export, When rendered, Then it is north-oriented, gridded, and laid out for print. [screenshot]

US-6.4 - Batch multi-page PDF export

- **AC-6.4.1** Given a building with multiple maps, When a staff member queues a batch export, Then an export job is created and a correctly ordered multi-page PDF is produced.
- **AC-6.4.2** Given a queued export job, When it fails mid-render, Then the job is marked failed and the partial output is not delivered as final.

US-6.5 - Z-axis floor labels

- **AC-6.5.1** Given a multi-floor facility, When a map and its markers are saved, Then each carries a z-axis floor label in its compliance metadata.

US-6.6 - Marker placement fidelity

- **AC-6.6.1** Given a marker placed at a canvas point, When the map is re-rendered and exported, Then the marker appears at the same coordinates with no drift or jitter. [screenshot]

E7 - Unified Scan Library & Media

US-7.1 - Unified media asset library

- **AC-7.1.1** Given a facility with mixed media, When the library opens, Then floorplan PDFs, point clouds, splats, walkthrough videos, and reference images all list in one view. [screenshot]

US-7.2 - Large multipart upload

- **AC-7.2.1** Given a large scan file, When uploaded via multipart, Then all parts assemble into one R2 object and the asset becomes available.
- **AC-7.2.2** Given a multipart upload interrupted partway, When resumed, Then it completes without re-uploading already-sent parts.

US-7.3 - Asset versioning

- **AC-7.3.1** Given an asset, When a new scan is uploaded for the same building, Then it is stored as a new version carrying capture date and surveyor, leaving prior versions intact. [screenshot]

US-7.4 - Point-cloud cold archive

- **AC-7.4.1** Given a master PLY point cloud, When ingested, Then it is written to the cold storage tier.
- **AC-7.4.2** Given a client request for an asset, When the asset is a master PLY, Then it is never served to the client.

US-7.5 – SPZ delivery format

- **AC-7.5.1** Given a scan for client delivery, When the viewer requests it, Then it is served as SPZ, not PLY.
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E8 – 3D Scan Viewer

US-8.1 – MP4 walkthrough player

- **AC-8.1.1** Given a facility with a walkthrough mp4, When a first responder opens it, Then the video plays with standard transport controls on the device. [screenshot]

US-8.2 – Interactive splat viewer

- **AC-8.2.1** Given an SPZ scan, When the splat viewer loads, Then the scene renders and responds to orbit, fly, and walk camera modes. [screenshot]
- **AC-8.2.2** Given a device without WebGL2, When the viewer loads, Then it shows a graceful fallback message rather than a broken canvas. [screenshot]

US-8.3 – 3D measurement tool

- **AC-8.3.1** Given a loaded scan, When a staff member picks two points, Then a point-to-point distance is reported in real-world units. [screenshot]

US-8.4 – Anchored 3D markers

- **AC-8.4.1** Given a scan, When a staff member drops a marker with a title, description, and saved viewpoint, Then it anchors at that 3D position and reloads in place. [screenshot]
- **AC-8.4.2** Given a scene at the marker cap, When another marker is added, Then the cap is enforced with a clear message.

US-8.5 – Floor switching

- **AC-8.5.1** Given a multi-floor facility, When a responder switches floors, Then the viewer loads the selected floor's scan and updates the active-floor indicator. [screenshot]

US-8.6 – Named viewpoints

- **AC-8.6.1** Given a scan, When a staff member saves a named viewpoint, Then selecting it later returns the camera to the stored position, target, and FOV. [screenshot]

US-8.7 – Scan version compare

- **AC-8.7.1** Given two scan versions, When a client reviewer compares them, Then both capture dates are shown and the viewer can swap between versions. [screenshot]
-

E9 – Collaboration

US-9.1 – Anchored comment threads

- **AC-9.1.1** Given a map region or 3D marker, When a client reviewer starts a thread, Then the comment anchors to that location and renders for all participants. [screenshot]
- **AC-9.1.2** Given an existing thread, When a participant replies, Then the reply nests under the thread in order.

US-9.2 – Resolve workflow

- **AC-9.2.1** Given an open thread, When a staff member resolves it, Then it is hidden from the default view but retained in the record, not deleted. [screenshot]

US-9.3 – @-mentions with deep-link email

- **AC-9.3.1** Given a comment, When a client reviewer @-mentions a staff member, Then that member receives an email containing a deep link to the comment.
- **AC-9.3.2** Given an @-mention of a user outside the org, When submitted, Then the mention is rejected and no email is sent.

US-9.4 – Batched activity notifications

- **AC-9.4.1** Given multiple activity events for one user within a batching window, When the digest sends, Then they arrive as a single batched notification.

US-9.5 – External share links

- **AC-9.5.1** Given a map or scan, When an operator admin creates a share link, Then an unauthenticated visitor with the link sees only that scoped resource. [screenshot]
 - **AC-9.5.2** Given a revoked share link, When it is opened, Then access is denied.
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E10 – Global Search

US-10.1 – Search across all entity types

- **AC-10.1.1** Given indexed content, When a staff member searches a term, Then matching facilities, buildings, projects, documents, and marker labels are returned grouped by type. [screenshot]

US-10.2 – Org-scoped search results

- **AC-10.2.1** Given a client reviewer in Org A, When they search a term that also matches Org B content, Then only Org A results are returned.

US-10.3 – Search-centric navigation for large lists

- **AC-10.3.1** Given a large set of facilities, When a staff member searches by name, Then a matching facility is reachable directly from results without hierarchy traversal. [screenshot]
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E11 – Compliance & Trust

US-11.1 – NG911/NENA GeoJSON export

- **AC-11.1.1** Given a facility with structure and point data, When a compliance officer exports GeoJSON, Then the output conforms to the NENA STA-006 schema.
- **AC-11.1.2** Given a facility missing required NENA fields, When export is attempted, Then the export reports the missing fields rather than producing invalid GIS.

US-11.2 – Compliance metadata per facility

- **AC-11.2.1** Given a facility, When a compliance officer edits compliance metadata, Then last-reviewed date, last responder tour, mandate flags, and drill links are saved. [screenshot]

US-11.3 – Immutable, exportable audit log

- **AC-11.3.1** Given audit entries, When a compliance officer exports the log, Then it contains who viewed, edited, and exported which map and when.
- **AC-11.3.2** Given an existing audit entry, When any actor attempts to alter it, Then the entry cannot be modified or deleted.

US-11.4 – Public trust page

- **AC-11.4.1** Given the deployment, When a visitor opens the trust page, Then it states the application-level controls and compliance posture for the active brand. [screenshot]

US-11.5 – Re-verification reminders

- **AC-11.5.1** Given a facility whose last-reviewed date has passed the staleness threshold, When the reminder check runs, Then the facility is flagged for re-verification. [screenshot]

E12 – Accessibility

US-12.1 – Full keyboard navigation

- **AC-12.1.1** Given the portal, When a user navigates with keyboard only, Then all interactive controls are reachable and operable with a visible focus indicator. [screenshot]

US-12.2 – Screen-reader labels

- **AC-12.2.1** Given the interface and map symbols, When inspected by a screen reader, Then every control and symbol exposes a descriptive accessible label.

US-12.3 – Non-visual map alternative

- **AC-12.3.1** Given a map, When a user opens its non-visual alternative, Then a text list of rooms, exits, and markers conveys the same information. [screenshot]

US-12.4 - Contrast and VPAT posture

- **AC-12.4.1** Given the UI, When color contrast is measured, Then text and controls meet WCAG 2.1 AA contrast ratios. [screenshot]
 - **AC-12.4.2** Given a procurement request, When the VPAT is requested, Then a maintained VPAT/ACR document is available.
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E13 - Admin & Platform Ops

US-13.1 - User management

- **AC-13.1.1** Given the admin area, When an operator admin creates a user and assigns a role, Then the user can log in with exactly that role's permissions. [screenshot]
- **AC-13.1.2** Given an active user, When an operator admin deactivates them, Then their next request is rejected.

US-13.2 - API key management

- **AC-13.2.1** Given the admin area, When an operator admin issues a scoped API key, Then the key authenticates API requests within its scope. [screenshot]
- **AC-13.2.2** Given an issued key, When the admin revokes it, Then subsequent requests using that key are rejected.

US-13.3 - Audit log viewer

- **AC-13.3.1** Given audit entries, When a compliance officer filters the in-portal viewer by actor and date, Then only matching entries are shown. [screenshot]

US-13.4 - Platform settings

- **AC-13.4.1** Given the settings area, When an operator admin changes a deployment-wide setting, Then the change persists and takes effect without a code change.

US-13.5 - Observability

- **AC-13.5.1** Given a runtime error in the deployment, When it occurs, Then it is captured in the observability output with enough context to diagnose it.
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E14 - Quality & Delivery Harness

US-14.1 - Failing test before implementation

- **AC-14.1.1** Given a new acceptance criterion, When its test `T-<AC>` is first written, Then it fails against the unimplemented feature before any implementation exists.

US-14.2 - Engine unit tests

- **AC-14.2.1** Given a ported engine, When its extraction lands, Then it ships with passing Vitest unit tests covering its core logic.

US-14.3 - Playwright screenshot assertions

- **AC-14.3.1** Given a UI-visible outcome, When its Playwright test runs, Then a screenshot is captured and compared against the stored baseline. [screenshot]
- **AC-14.3.2** Given a visual regression, When the screenshot differs beyond threshold, Then the test fails.

US-14.4 - CI on every change

- **AC-14.4.1** Given a change, When it is pushed, Then CI runs the full Vitest and Playwright suite and blocks merge on failure.

US-14.5 - Deploy to isolated CF test project

- **AC-14.5.1** Given a deploy, When `wrangler pages deploy` runs, Then it targets the `pathfinder` Pages project and never `els911-portal` production.

Coverage summary

Epic	Title	#Stories	#ACs
E1	White-Label Brand Layer	5	8
E2	Identity, Roles & Access	5	7
E3	Org Hierarchy & Dashboards	5	7
E4	Project & Review Workflow	5	8
E5	2D Floorplan Annotation Engine	5	7
E6	Safety Mapping & NFPA Export	6	8
E7	Unified Scan Library & Media	5	7
E8	3D Scan Viewer	7	9
E9	Collaboration	5	8
E10	Global Search	3	3
E11	Compliance & Trust	5	7
E12	Accessibility	4	5
E13	Admin & Platform Ops	5	7
E14	Quality & Delivery Harness	5	6
Total		70	97